

TABLE VII
(CONTINUED)

	1. 1970– 1980 extensive margin	2. 1980– 1990 extensive margin	3. 1990– 1998 extensive margin	4. 1970– 1998 extensive margin	5. 1980– 1998 extensive margin	6. 1980– 1998 intensive margin	7. 1980– 1998 extensive + intensive
D. 10× annual changes in college-equivalent share of employment in percentage points, predicted by impact of computerization on task input (panel B)							
	NIPA computer investment measure				CPS computer use measure		
Nonroutine tasks	0.40	0.65	1.10	0.69	1.41	0.40	1.81
Routine tasks	0.29	0.48	0.81	0.51	0.80	1.04	1.84
All tasks	0.70	1.12	1.91	1.19	2.21	1.44	3.65
E. Estimated log demand shifts for college-equivalent/noncollege-equivalent labor 1970–1998 (100 × annual log changes)							
	Using constant-elasticity of substitution model to estimate changes in college demand						
$\sigma = 0.0$	4.99	2.53	2.25	3.33			2.41
$\sigma = 1.4$	3.95	4.65	2.76	3.86			3.81
$\sigma = 2.0$	3.50	5.56	2.98	4.09			4.41
	Using task model to predict changes in college demand						
Total task Δ (panel C)	1.23	1.29	1.43	1.31	1.56	−0.06	1.51
Predicted by computerization (panel D)	0.64	0.70	0.98	0.76	1.39	0.91	2.29

Panels C and D: Implied employment share changes for college-equivalent labor (in percentage points) are calculated as the inner product of observed or predicted changes in task input from panels A and B and the coefficient vector from a fixed coefficient model of educational input. For extensive margin task shifts, this coefficient vector is estimated from a regression of college equivalent employment (in FTEs) in 140 consistent CIC industries on the five DOT measures of industry task input (in centiles) and a constant using the 1980 MORG sample. For intensive margin task shifts, the coefficient vector is estimated from a regression of college equivalent employment in 470 COC occupations on the five DOT measures of occupational task input (in centiles) and a constant using the 1984 CPS sample. College-equivalents labor is defined as all workers with college or greater education plus half of those with some college.

Panel E: Fixed coefficients log relative demand shifts are calculated as the change in the log ratio of college-equivalent/noncollege-equivalent employment using the initial (1970, 1980, or 1990) college-equivalent/employment share in full-time equivalents and the implied percentage point change in this share from Panels C and D.

Constant Elasticity of Substitution (CES) implied relative demand shifts for college-equivalent labor are calculated following Autor, Katz, and Krueger [1998] using a CES aggregate production function with two inputs, college and high school equivalents, and an elasticity of substitution denoted by σ . See Table II of Autor, Katz, and Krueger for details.

We next estimate the contribution of task changes attributable to computerization to growth in college graduate employment. These estimates, shown in panel D, implement equation (18) using the within-industry and within-occupation task shifts predicted by computerization (panel B). The estimates of the “computer-induced” changes in college employment are comparable in magnitude to, and in some cases larger than, the analogous estimates based on observed task inputs. We find, for example, that computer-induced task changes along the extensive margin contributed 1.2 percentage points per decade to growth in college employment during 1970 to 1998, and 2.2 percentage points during 1980 to 1998. Adding intensive margin task changes raises this estimate considerably. In net, we estimate that task shifts attributable to computerization increased college employment by 3.7 percentage points per decade from 1980 to 1998.

The final step of our estimation is to benchmark employment changes induced by shifting task demands against conventional estimates of demand for college labor calculated for the same period. For this benchmarking exercise, we use the familiar constant elasticity of substitution (CES) framework with two factors of production—college equivalent and high school equivalent labor—to estimate log relative demand shifts favoring college labor during 1970 to 1998. Under the assumptions that the economy operates on the demand curve and that factors are paid their marginal products, the CES model calculates the implied shift in college/noncollege relative demand consistent with observed shifts in relative employment and earnings of college versus noncollege workers.²⁸ This procedure also requires us to assume an elasticity of substitution, σ , between college and high school equivalent workers. Following a large literature, we use values of σ , ranging from 0 to 2, with a consensus estimate of $\sigma = 1.4$.

Panel E of the table presents the benchmark CES estimates for the period 1970 to 1998. We calculate that relative demand for

28. The demand model is $Q_t = [\alpha_t(a_t N_{ct})^\rho + (1 - \alpha_t)(b_t N_{ht})^\rho]^{1/\rho}$, where Q is aggregate output, N_{ct} , N_{ht} are quantities of employed college and high school equivalent labor, a_t , b_t are factor-augmenting technological parameters, α is an index of the share of work activities allocated to college versus high school labor, and the elasticity of substitution is given by $\sigma = 1/(1 - \rho)$. The demand index is $D_t = \sigma \ln(\alpha_t/[1 - \alpha_t]) + (\sigma - 1)\ln(a_t/b_t)$. Katz and Murphy [1992], Johnson [1997], Murphy, Romer, and Riddell [1999], and Acemoglu [2002] implement similar models. Our estimates are based on Table II of Autor, Katz, and Krueger [1998] and updated in 1998 (from 1996) for this analysis. Unlike these authors, we also include estimates for $\sigma = 0$ since our fixed coefficients model incorporates this assumption.

college-educated labor grew rapidly between 1970 and 1998. Using the consensus value of $\sigma = 1.4$, we estimate that demand for college labor rose by 3.9 log points annually. If we instead assume that $\sigma = 0$, we obtain a smaller—though still quite rapid—demand shift of 3.3 log points annually.²⁹

In the bottom of panel E we finally compare our task-based demand estimates with the CES-based numbers. To facilitate comparison, we convert predicted changes in the college employment level (panels C and D) into changes in log relative college-equivalent/noncollege employment. As noted above, this task-based calculation ignores wage changes and hence corresponds to an elasticity of substitution of $\sigma = 0$. It is therefore useful to compare the task-based demand estimates to the corresponding CES numbers using both the consensus elasticity of $\sigma = 1.4$ and the more comparable, albeit less realistic, value of $\sigma = 0$.

As is visible in panel E, the task model explains a sizable share—25 to 65 percent—of the estimated growth in college-equivalent/noncollege-equivalent demand in each decade. Consistent with the accelerating rate of task change shown in Figure I, the smallest share of the overall demand shift is explained by task shifts in the 1970s and the largest share in the 1990s.

Comparing task changes potentially attributable to computerization to the CES demand index for these three decades, we find that these extensive margin task changes explain 20 to 25 percent of the estimated demand shift for college versus noncollege labor during 1970 to 1998. If we focus on only the two most recent decades and include both intensive and extensive margin changes, the task model can explain a large fraction—60 to 90 percent—of the estimated increase in relative demand for college employment. Notably, almost 40 percent of the computer contribution to rising educational demand in the last two decades is due to shifts in task composition within nominally unchanging occupations.

In net, these illustrative calculations demonstrate that changes in task demands accompanying workplace computerization are economically large and—with caveats noted—could have contributed substantially to relative demand shifts favoring educated labor in the United States since 1970.

29. The estimated demand shift is smaller in the latter calculation because a higher value of σ places greater weight on relative wage changes, and the relative earnings of college graduates rose rapidly after 1980.

VII. CONCLUSION

What is it that computers do—or what is it that people do with computers—that appears to increase demand for educated workers? This paper formalizes and tests an intuitive answer to this question that has been informally articulated by scholars in a number of disciplines over several decades. Computer technology substitutes for workers in performing routine tasks that can be readily described with programmed rules, while complementing workers in executing nonroutine tasks demanding flexibility, creativity, generalized problem-solving capabilities, and complex communications. As the price of computer capital fell precipitously in recent decades, these two mechanisms—substitution and complementarity—have raised relative demand for workers who hold a comparative advantage in nonroutine tasks, typically college-educated workers.

Our task framework emphasizes that the causal force by which advancing computer technology affects skill demand is the declining price of computer capital—an economywide phenomenon. We developed a simple model to explore how this price decline alters task demand within industries and occupations. This model predicts that industries that were intensive in labor input of routine tasks in the precomputer era would make relatively larger investments in computer capital. Simultaneously, they would reduce labor input of routine tasks, for which computer capital substitutes, and increase demand for nonroutine task input, which computer capital complements.

Employing consistent, representative, time series observations on the task composition of jobs from the *Dictionary of Occupational Titles*, we affirm these predictions across several margins of task change and estimate that they may have contributed substantially to demand shifts favoring educated labor over the past three decades. We also considered several alternative explanations for our findings, most significantly the rising human capital and labor force attachment of women. We find that the documented task shifts, and their associations with the adoption of computer technology, are as evident *within* gender, education, and occupation groups as between them. The pervasiveness of these shifts suggests to us that changes in job task content—spurred by technological change—may plausibly be viewed as an underlying factor contributing to recent demand shifts favoring educated labor.